A-A-54205
25 June 1990

COMMERCIAL ITEM DESCRIPTION

STEREOSCOPE, VISION TESTING

The General Services Administration has authorized the use of this Commercial Item Description.

This Commercial Item Description covers a portable, self-enclosed, table top model, stereoscopic vision tester designed and constructed to test for deficiencies in visual acuity, muscle balance, fusion and depth perception at near and far distances.

Salient Characteristics:

SYSTEM REQUIREMENTS.
The vision tester shall meet all requirements of UL 544, Standard for Safety of Medical and Dental Equipment, as determined and tested by Underwriters Laboratory Incorporated. The unit shall be UL listed and bear the appropriate UL listing mark.

The vision tester shall weigh no more than 30 pounds and require a maximum of three (3) square feet of surface area, suitable for table top use.

The viewing lenses shall be molded, ground and polished of ophthalmic quality glass.

The viewing position shall be adjustable to accommodate at least persons ranging in height from three (3) feet to seven (7) feet tall while in a sitting position.

The housing shall be constructed of flame retardant, non-conductive, high impact strength ABS material or an alternate material which shows the same or better mechanical properties.

The unit shall be designed to accommodate a person wearing spectacle frames and will not permit exterior ambient light to interfere with tests. Only the applicant being screened or the administrator can view the test being presented.

The unit shall be designed such that far and near point tests are generally conducted in the normal viewing axis. Testing of bifocal wearers shall not require excessive head movement or repositioning of the apparatus to transition from near to far point tests.

AMSC N/A FSC 6540
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A brow/forehead rest shall be incorporated into the design that use disposable rest tissues to ensure proper hygienic conditions.

A pointer shall be supplied and an observation/training portal shall be incorporated into the design of the vision tester that will permit easy access to targets for assistance in test administration.

The unit shall have a manual test advance mechanism that will permit testing in either a forward or reverse direction. An automatic (electronic) test advance mechanism with a manual over-ride is also acceptable. The advance mechanism shall include a position/test indicator that will accurately identify the current test in position for viewing. The manufacturer is responsible to identify in the operating instructions the correlation between the indicator test position and test number.

The instrument shall have a name plate securely attached in plain view on its external surface. The name plate shall be legibly and permanently marked with at least the following information:

a. Name or registered trademark of the manufacturer.
b. Type or model number.
c. Serial number.

The unit shall operate from a nominal 115 VAC, 50/60 Hz power supply.

ILLUMINATION.
The power input to the unit shall be controlled through a main switch located on the unit itself. In addition, an automatic illumination activator shall be incorporated into the design that will automatically activate the main illumination only when the test subject is in proper viewing position. A manual over-ride shall be included which, when desired, will defeat the automatic illumination activator.

All test targets shall be transilluminated.

The illumination source shall be evenly diffused throughout the entire viewing area.

The emitted white background light shall render daylight conditions at a nominal color temperature of 5000 degrees Kelvin. The contrast level between an opaque target and the background shall be a minimum of 98 %.

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The luminance at the viewers point shall be 10-15 foot-lamberts.

All lamps shall be rated at a minimum of 1000 hours lab life and shall be easily accessible for replacement by field personnel.

All internal components within the viewing field shall be appropriately finished to minimize internal reflection.

All internal light sources shall be completely shielded from the viewing field as not to interfere with testing.

TESTS.
The content of targets/optotypes are proprietary for sole use by the Department of Defense. Their content shall not be made available to the general public.

The tester shall permit testing of each eye individually (monocular) and both eyes simultaneously (binocular). The internal design shall protect the occluded eye from receiving any light from the unoccluded target.

The test targets/optotypes shall be presented through the use of slides. One complete set of test slides shall be provided for each unit. Test slides shall be easily removed for cleaning and replacement by field personnel.
The specification for test slides and the targets/optotypes shall be specified in the contract or purchase order. See ordering data.

The vision tester shall at least perform the following 17 tests and each test shall be identified by the numbers to their left. At a minimum, the following nine (9) tests shall be installed in the machine simultaneously and may be performed without removal or replacement of slides:

FAR TESTS	NEAR TESTS
1. Vertical Phoria	6. Vertical Phoria
2. Lateral Phoria	7. Lateral Phoria
3. Monocular Acuity (small letters)	8. Monocular Acuity (small letters)
4. Monocular Acuity (large letters)	9. Monocular Acuity (large letters)
5. Fusion and Depth Perception	

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In addition, the unit shall perform the following optional eight (8) tests which may be stored outside of the viewer itself. These tests can be administered by removal and replacement of slides.

FAR TESTS	NEAR TESTS
3A. Alternate Monocular Acuity (small letters)	13. Binocular Acuity (small letters)
4A. Alternate Monocular Acuity (large letters)	14. Binocular Acuity (large letters)
5A. Alternate Fusion and Depth Perception	
10. Landolt Ring Acuity	
11. Binocular Acuity (small letters)	
12. Binocular Acuity (large letters)	

A reusable container designed to hold and protect any slides held outside of the viewer during extended periods of non-use shall be supplied with each unit.

Working distances for far tests shall be 20 feet to infinity and near test 13-14 inches.

ACCESSORIES.
Each unit shall be provided with the following accessories:

* 1 maintenance manual.
* 1 operating manual.
* 100 record forms.
* 500 clean-rest tissues.
* 2 sets of replacement bulbs.
* 1 dust cover.
* 1 plastic laminated 8-1/2 by 11 scoring key.
* 1 replacement parts list including a list of replacement slides.
* 1 demonstration apparatus for depth perception.

The operators manual shall at a minimum include the following contents:

a) Description of Instrument and features.
b) Administration of tests; shall include the information as outlined in the Manual of Instructions: Armed Forces Vision tester NSN 7610-00-721-9390 as modified to their particular instrument.
c) A scoring key.
d) Glossary.

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The depth perception demonstration apparatus shall be a clear translucent plastic block attached to the tester, with nominal dimensions of 1-1/2" wide by 4" long by 1/2" thick. The block shall contain 5 circles, spaced evenly apart across the length of block. 4 circles shall be printed on the back face and the center circle shall be printed on the front face. See Figure 1 for illustration.

Workmanship. The vision tester shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One vision tester, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

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Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings. specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented be Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document.

NATIONAL STOCK NUMBER	ITEM IDENTIFICATION
6540-00-299-8048	STEREOSCOPE, VISION TESTING

Ordering data. (Test slides and targets/optotypes designs, intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.)

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MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
PGC 51193

Project Number: 6540-1248

Location: ENABLE/CSTEREO/S35/1

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